THE NEWLY DEFINED SHRIEK FUNCTORS FORM A PSEUDO FUNCTOR AND ARE COMPATIBLE WITH THE DUALITY INVOLUTION AND WITH THE RIGHT ADJOINT FOR A PROPER MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AUE (sources/stacks-duality/pinned-original-source.tex line 5588).

Allowed standard prerequisites (editorial).

- The duality involution on the bounded coherent derived category. Same chapter, Lemma labelled lemma-dualizing-schemes, cited at proof lines 5640-5644 and 5668. Supplies eta_Y, eta_Z and part (3) of the statement.
- The upper shriek through duality and the right adjoint. Same chapter, Lemma labelled lemma-shriek-via-duality, cited at proof lines 5674-5676, 5678-5680 and 5684-5686. Identifies f^!_new with the right adjoint for the proper case and with f^! for the separated case.
- Identification of the good dualizing complex with the shriek of the base complex for a proper morphism. Same chapter, Lemma labelled lemma-proper-map-good-dualizing-complex, cited at proof lines 5670-5674. Gives omega_X^bullet = a(omega_Y^bullet) needed to change the base.
- The Noetherian example comparing the two internal Hom expressions. Same chapter, Example labelled example-iso-on-RSheafHom-noetherian, cited at proof lines 5682-5684. Supplies the displayed equalities of part (6).

Source reference. Original reference 0AU4: sources/stacks-duality/pinned-original-source.tex, source lines 5259–5259. Same-chapter pointer into the author environment situation at source line 5259; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A89: sources/stacks-duality/pinned-original-source.tex, source lines 472–472. Same-chapter pointer into the author environment

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lemma at source line 472; invoked at proof line 5649 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 003F: sources/stacks-chow/tags.txt, source lines 132–132. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5650 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 5668 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AU2: sources/stacks-duality/pinned-original-source.tex, source lines 4734–4734. Same-chapter pointer into the author environment lemma at source line 4734; invoked at proof line 5673 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AUD: sources/stacks-duality/pinned-original-source.tex, source lines 5541–5541. Same-chapter pointer into the author environment lemma at source line 5541; invoked at proof line 5676 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0GEW: sources/stacks-duality/pinned-original-source.tex, source lines 865–865. Same-chapter pointer into the author environment example at source line 865; invoked at proof line 5681 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $(S, \omega_S^\bullet)$ be as in Situation 0AU4. With $f^!_{new}$ and $\omega_X^\bullet$ defined for all (morphisms of) schemes of finite type over S as above:*

1. *the functors $f^!_{new}$ and the arrows $(g \circ f)^!_{new} \to f^!_{new} \circ g^!_{new}$ turn D^+_{Coh} into a pseudo functor from the category of schemes of finite type over S into the 2-category of categories,*
2. *$\omega_X^\bullet = (X \to S)^!_{new}\omega_S^\bullet$,*
3. *the functor D_X defines an involution of $D_{Coh}(\mathcal{O}_X)$ switching $D^+_{Coh}(\mathcal{O}_X)$ and $D^-_{Coh}(\mathcal{O}_X)$ and fixing $D^b_{Coh}(\mathcal{O}_X)$,*
4. *$\omega_X^\bullet = f^!_{new}\omega_Y^\bullet$ for $f : X \to Y$ a morphism of finite type schemes over S,*
5. *$f^!_{new}M = D_X(Lf^*D_Y(M))$ for $M \in D^+_{Coh}(\mathcal{O}_Y)$, and*
6. *if in addition f is proper, then $f^!_{new}$ is isomorphic to the restriction of the right adjoint of $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$ to $D^+_{Coh}(\mathcal{O}_Y)$ and there is a canonical isomorphism*

$$Rf_*R\mathcal{H}om_{\mathcal{O}_X}(K, f^!_{new}M) \to R\mathcal{H}om_{\mathcal{O}_Y}(Rf_*K, M)$$

for $K \in D^-_{Coh}(\mathcal{O}_X)$ and $M \in D^+_{Coh}(\mathcal{O}_Y)$, and

$$Rf_*R\mathcal{H}om_{\mathcal{O}_X}(K, \omega_X^\bullet) = R\mathcal{H}om_{\mathcal{O}_Y}(Rf_*K, \omega_Y^\bullet)$$

for $K \in D^-_{Coh}(\mathcal{O}_X)$ and

If X is separated over S, then $\omega_X^\bullet$ is canonically isomorphic to $(X \to S)^!\omega_S^\bullet$ and if f is a morphism between schemes separated over S, then there is a canonical isomorphism[1] *$f^!_{new}K = f^!K$ for K in D^+_{Coh}.*

Proof. Let $f : X \to Y$, $g : Y \to Z$, $h : Z \to T$ be morphisms of schemes of finite type over S. We have to show that

$$\begin{array}{ccc} (h \circ g \circ f)^!_{new} & \longrightarrow & f^!_{new} \circ (h \circ g)^!_{new} \\ \downarrow & & \downarrow \\ (g \circ f)^!_{new} \circ h^!_{new} & \longrightarrow & f^!_{new} \circ g^!_{new} \circ h^!_{new} \end{array}$$

is commutative. Let $\eta_Y : \mathrm{id} \to D_Y^2$ and $\eta_Z : \mathrm{id} \to D_Z^2$ be the canonical isomorphisms of Lemma 0A89. Then, using Categories, Lemma 003F, a computation (omitted) shows that both arrows $(h \circ g \circ f)^!_{new} \to f^!_{new} \circ g^!_{new} \circ h^!_{new}$ are given by

$$1 \star \eta_Y \star 1 \star \eta_Z \star 1 : D_X \circ Lf^* \circ Lg^* \circ Lh^* \circ D_T \longrightarrow D_X \circ Lf^* \circ D_Y^2 \circ Lg^* \circ D_Z^2 \circ Lh^* \circ D_T$$

This proves (1). Part (2) is immediate from the definition of $(X \to S)^!_{new}$ and the fact that $D_S(\omega_S^\bullet) = \mathcal{O}_S$. Part (3) is Lemma 0A89. Part (4) follows by the same argument as part (2). Part (5) is the definition of $f^!_{new}$.

Proof of (6). Let a be the right adjoint of Lemma 0A9E for the proper morphism $f : X \to Y$ of schemes of finite type over S. The issue is that we do not know X or Y is separated over S (and in general this won't be true) hence we cannot immediately apply Lemma 0AU2 to f over S. To get around this we use the canonical identification $\omega_X^\bullet = a(\omega_Y^\bullet)$ of Lemma 0AUD. Hence $f^!_{new}$ is the restriction of a to $D^+_{Coh}(\mathcal{O}_Y)$ by Lemma 0AU2 applied to $f : X \to Y$ over the base scheme Y! The displayed equalities hold by Example 0GEW.

The final assertions follow from the construction of normalized dualizing complexes and the already used Lemma 0AU2. □

[1] We haven't checked that these are compatible with the isomorphisms $(g \circ f)^! \to f^! \circ g^!$ and $(g \circ f)^!_{new} \to f^!_{new} \circ g^!_{new}$. We will do this here if we need this later.